Sputtering-Annealing

Phase 02 | Operator explanation and control logic

Release	Build	Reviewed	Executable
0.9.41	2026.09.04-r20	2026-09-04	Phase 02

Safety boundary: this guide describes the software sequence only. Follow the current laboratory SOP, confirm the physical chamber and instruments, and remain supervised. Software limits do not replace hardware interlocks or the emergency procedure.

Purpose

Run the supervised COSCON sputtering and oven-annealing sequence for the configured number of cycles while keeping the argon leak valve under manual operator control.

Scope and ownership

The phase uses direct COSCON UDP control and an operator-focused dashboard. COSCON background polling is coordinated with activation and Reset transactions so status reads cannot interleave the target handoff.

Operator sequence

1	Confirm the sputter-gun/electronics pre-flight and verify the physical argon valve is closed before starting.
2	Start Phase 02 and complete any dashboard prompts. The default recipe performs one COSCON Degas before cycle 1.
3	When prompted, open the argon leak valve manually and wait for the pressure guidance window; keep physical pressure supervision active.
4	The controller pauses background polling, validates 2250 V / 10.0 mA, waits 1.5 s, switches to Operate and verifies mode, interlock, energy, emission and pressure.
5	Allow sputtering to run for 20 min per cycle. Five consecutive valid target readings are required before the timed step begins.
6	When prompted, return COSCON to Standby and close the argon leak valve manually before the anneal sequence proceeds.
7	The oven PID is driven to 620 deg C, held for 10 min inside its stability window, then reset to 0 deg C after the cycle.
8	Repeat for all cycles. Use Abort / Safe Stop whenever the physical condition is uncertain or a prompt cannot be satisfied.

Packaged defaults

Parameter	Default	Meaning
Cycles	3	Sputtering-annealing repetitions.
Initial Degas	Enabled	One automatic Degas before cycle 1.
Sputtering hold	20 min/cycle	Timed only after target qualification.
COSCON target	2250 V / 10.0 mA	Energy and emission target.
Stable output	5 valid reads	Consecutive energy/emission confirmation.
Argon target	2.0e-5 mbar	Operator pressure guidance target.
Pressure warning	3.0e-5 mbar	Software warning threshold.
Pressure emergency	1.0e-4 mbar	Software emergency stop threshold.
Anneal	620 deg C / 10 min	Oven target and in-band hold.
PID reset	0 deg C	Setpoint written after every anneal and on safe stop when configured.

COSCON activation recovery

Only the exact pre-sputter HV-Module Energy Overload condition is recoverable. The default policy allows one verified eight-second retry after checking the safe inactive state, pressure and interlock. A repeated exact overload may trigger one guarded Reset recovery.

Reset is allowed only with the manual valve closed, output de-energized, interlock acceptable and background polling paused. Phase 02 waits up to 60 s for reconnection, requires three consecutive Off / Interlock=Ok / near-zero readings, proves the reboot, reopens the valve only after the operator prompt and repeats 60 s of pressure conditioning before one final activation attempt. Arbitrary errors, unsafe output or a further overload are fatal.

> Never treat a pressure warning, COSCON status label or dashboard color as proof that the physical chamber is safe. Inspect the chamber, valve, cabling and interlocks and follow the laboratory response procedure.

Finish and abort behavior

Normal completion	Abort / Safe Stop
After the final anneal, the controller resets the PID, completes the final files and returns the instruments to the documented phase-end state.	Abort leaves COSCON and the oven in their safe-stop path, closes software resources and records the reason. The operator remains responsible for the physical valve and chamber state.

Data written

A new folder is reserved below Data Samples/Sputtering-Annealing Data using the pattern Sputtering-Annealing <sample> data NN.

Important record	Use
sputter_anneal_log.csv	Fixed-width telemetry for pressure, oven, Keysight and COSCON energy/emission/filament/anode/repeller values.
Action/event log	Manual prompts, valve transitions, activation checks, recovery decisions and safe-stop reasons.
Effective parameters	Actual recipe, pressure thresholds and recovery limits used by the run.